AT-03

Sk: Ashwaq Ahamed

Reg No: 192572070

Sub: Data structure

Code: CSA0306

1.

Linked List operation

Problem understanding

A linked list is literally empty. perform the given operations and show the final configuration.

(a) operations.

- L1. append(10)
- L1 . append(20)
- L1 . append(15)

Final linked list.

Head → 10 → 20 → 15 → NULL.

(b) operations

- L2. append(10)
- L2 . append(20)
- L2. append(15)
- L2. movetostart()
- L2. insert(39)
- L2. next()
- L2 . insert(12)

After append

10 → 20 → 15.

After To start() & insert(39).

39 → 10 → 20 → 15.

After next() & insert(12):

39 → 12 → 10 → 20 → 15 → Null

Final Answer

Head → 39 → 12 → 10 → 20 → 15 → Null

Time Complexity :- Append O(1), insert - O(1)

2. # Binary search

Elements

66, 111, 212, 323, 432, 543, 643, 649, 777, 888

Case 1 Key = 111

- medium = 432
- Search left
- middle = 111
- Element found
- Position = 2 (index = 1)

Case 2 Key = 888

- middle = 432
- search Right.
- middle = 649
- search right.
- middle = 777
- Search Right.
- middle = 888
- Element found

Position = 10 (index = 9)

Case 3 Key = 999

- middle = 432
- Right
- middle = 649
- Right
- middle = 777
- Right.
- middle = 888
- Right.
- Element not found.

Complexity

- Best case = O(1), Average case = O(log n), worst case = O(log n)

Stack operation

Array size = 5.

operations

Push (45)
Push (22)
Push (11)
Push (77)
Pop()
Push(44)
Push (33)
Pop(), Pop(), Pop()

Execution

Push 45 → [45]

Push 22 → [45, 22]

Push 11 → [45, 22, 11]

Push 77 → [45, 22, 11, 77]

Pop → Remove 77

stack
[45, 22, 11]

Push 44
stack : [45, 22, 11, 48]

Push 33.

stack [45, 22, 11, 44, 33]

POP → Remove (33)

stack : [45, 22, 11, 44]

pop → Remove [44]

stack : [45, 22, 11]

pop → Remove [11]

stack → [45, 22]

TOP → 22 45

Topelement = 22

Index Position = 1

Time complexity : push = O(1), Pop = O(1)

Q4

Prefix & Postfix Expression

Expression

((a+b)+c* (d+e)+ f)* (g+h)

Charecter	Stack	Postfix
(	(	
(	((	
a	((	
+	((+	a
b	((+	a
+	((+	ab
c	((+	ab+
*	((+*	ab+c
(	((+*(	ab+c
d	((+*(	ab+c
+	((+*(+	ab+cd
e	((+*(+	ab+cd
)	((+*	ab+cde
+	((+*+	ab+cde+
f	((+	ab+cde+*+
)	(*	ab+cde+*+f
*	((*	ab+cde+*f+
(	((*	ab+cde+*f+
g	((+	ab+cde+*+f+g
+	((*+	ab+cde+*f+g
h	((*+)	ab+cde+*+f+gh
)		ab+cde+*+f+gh

Final Post fix Expression

ab+cde +* +f +gh +*

5. Queue Based Ticketing system

Problem understanding

A queue based ticketing system serves customers on arrival order. vip customer should receive priority service.

Data structure used

- simple Queue (FIFO)
- priority queue.

Comparision

simple queue	priority Queue
First come first serve.	Highest priority first
fair to everyone.	vip customers served first.
Equal waiting time.	vip waiting time is less.
Enqueue O(1), Dequeue O(1).	Insert / Delete O(log n)
Easy to implement.	more complex.

Advantages

- Faster service for vip customer.
- suitable for hospital, airport, emergency service and premium customer service.

Disadvantage

- Regular customer wait longer
- less fair than FIFO
- Higher computational complexity